

Autonomous Job Application Agent

1. Description

The Autonomous Job Application Agent is a multi-agent AI pipeline that automates the full job application workflow. Given a job description and a candidate’s resume, the system researches the target company, extracts structured role requirements, and generates three personalised outputs in parallel: a tailored resume, a cover letter, and an interview preparation guide.

A dedicated quality-control agent scores all outputs and either auto-retries generation (up to 3 times) or routes to a human-in-the-loop review gate, where the candidate can approve or provide feedback for further refinement. Approved documents are exported as markdown files.

Framework LangGraph 0.2+ / LangChain	LLM GPT-4o (OpenAI)	Pattern Multi-agent DAG + HITL
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2. Architecture Overview

The system is a LangGraph **StateGraph** — a directed graph where each node is a specialised agent. All agents share a single typed state object (**AgentState**) and communicate exclusively by reading from and writing partial updates to it. The graph has four execution tiers:

Tier	Nodes	Pattern
Entry Pipeline	parse_input → research_company → analyze_jd	Sequential — each node builds context for the next; web_search tool used for company intel
Generation	tailor_resume, write_cover_letter, prepare_interview	Parallel fan-out — all three execute concurrently, each writing to isolated state keys
Quality Gate	aggregate_outputs	Fan-in — LLM critic scores all docs (0-1); auto-retries if score < 0.7 (max 3×)
HITL Review	human_review	Interrupt — graph pauses, human approves or gives feedback, graph resumes via MemorySaver checkpoint

3. Multi-Agent DAG — Execution Flow

The diagram below shows the full graph topology: the orchestrator delegates to specialized sub-agents, each bound to its own tools, with results fan-in to a shared output/review node. Dashed arrows represent the conditional retry loop back through the graph.

